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Optimizing machine learning models for obesity risk prediction through hyperparameter tuning

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ABSTRACT

Early risk prediction is essential as obesity is a major public health concern associated with numerous chronic conditions. This study evaluates the effectiveness of various machine learning algorithms in predicting obesity risk with a particular focus on Hyperparameter optimisation enhances model performance. Obesity-related data was analysed using several algorithms including Decision Tree (DT), Support Vector Machine (SVM), K-Nearest Neighbours (KNN), and Logistic Regression (LR). Experimental results showed that Logistic Regression achieved the highest accuracy (99.63%), while Decision Tree produced the lowest (79.68%). Performance of SVM (89.6%) and KNN (94.33%) fell in between. Hyperparameter tuning significantly improved these models leading to greater robustness and predictive accuracy. Furthermore, a novel framework combining feature engineering with a cuckoo-inspired optimisation model was proposed to further enhance prediction quality. In addition, a novel hybrid framework integrating cuckoo-inspired feature optimization with Hyperparameter-tuned machine learSning models is proposed. This joint optimization strategy significantly enhances prediction accuracy, robustness, and model interpretability, distinguishing the proposed approach from existing obesity risk prediction methods.

1. Introduction

Obesity is rising at an alarming rate across both developed and developing nations making it one of most urgent global public health challenges [1]. World Health Organization (WHO) defines obesity as an abnormal or excessive accumulation of body fat that can be detrimental to health. Since 1975 global obesity rates have nearly tripled, with more than 1.9 billion people now classified as overweight and over 650 million of them as obese [2]. Rapid increase among children and adolescents is especially concerning. Beyond its health risks, obesity also carries profound economic and social costs including higher healthcare expenditures, reduced quality of life and an elevated risk of premature death [3]. Obesity is not only a disease itself but also a major risk factor for numerous chronic conditions. It is strongly linked to cardiovascular disease, hypertension, dyslipidaemia, type 2 diabetes, osteoarthritis, sleep apnoea and several cancers [4]. Additionally, it contributes to psychological challenges such as depression, low self-esteem, and social stigma. Causes of obesity are highly complex, stemming from interaction of genetic predisposition, lifestyle choices, eating behaviours, socio-economic status and environmental influences. Because of this complexity early identification of individuals at high risk is crucial for timely intervention, personalised treatment and effective public health strategies [5]. Conventional methods of estimating obesity risk typically rely on anthropometric measures such as body mass index (BMI), waist circumference and skinfold thickness along with demographic and lifestyle information. While useful these methods cannot fully capture nonlinear and multifactorial relationships among the diverse factors involved. Similarly, traditional statistical models such as linear regression are limited by assumptions of independence and linearity which may not hold in real-world health data [6]. These highlights need for more advanced predictive approaches capable of integrating heterogeneous datasets and uncovering hidden patterns. Machine learning (ML) has emerged as a powerful approach for prediction and decision-making in healthcare. ML algorithms are well-suited to handling complex datasets and detecting subtle interactions that traditional methods may overlook. Their adaptability improving through iterative training with larger datasets makes them particularly valuable in addressing evolving healthcare challenges. In obesity research, ML models can integrate diverse data sources, including dietary habits, lifestyle surveys, physiological measures, and even genetic markers, to provide more accurate

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risk predictions [7].

Among the most widely studied ML algorithms are Decision Trees (DT), Support Vector Machines (SVM), K-Nearest Neighbours (KNN), and Logistic Regression (LR). Each has distinct strengths and limitations. For instance, Decision Trees are interpretable but prone to overfitting, particularly with noisy or small datasets. SVMs are effective in high-dimensional classification tasks but require careful parameter tuning. KNN is simple and effective but computationally expensive with large datasets. Logistic Regression, while relatively simple, remains widely used because of its interpretability and strong performance in binary classification tasks such as obesity risk assessment [8]. Performance of these algorithms depends heavily on Hyperparameter tuning adjusting external configurations that control how models learn. Poorly tuned models may either underfit, missing key patterns, or over fit, performing well on training data but poorly on unseen data. Optimisation methods such as grid search, random search, Bayesian optimisation, and meta-heuristic algorithms are commonly employed to determine optimal [9]. Feature engineering is equally critical in developing effective models. By transforming raw data into meaningful features researchers can enhance a model's ability to detect relevant patterns. However, feature selection and transformation are often complex and computationally demanding. Nature-inspired optimisation methods have recently been explored to address this challenge, as they provide efficient strategies for searching high-dimensional feature spaces [10]. One promising example is cuckoo search algorithm which balances exploration and exploitation based on brood parasitism behaviour of cuckoo birds. This approach has shown success in improving classification accuracy by optimising both feature subsets and model parameters. This paper evaluates predictive performance of four widely used ML algorithms Decision Tree, Support Vector Machine, K-Nearest Neighbours and Logistic Regression in assessing obesity risk, with a particular focus on role of Hyperparameter optimisation. Experimental results show that Logistic Regression achieved highest accuracy (99.63 %), followed by KNN (94.33 %) and SVM (89.6 %) while Decision Tree performed worst (79.68 %). These findings underscore importance of parameter tuning for maximising predictive accuracy and generalisability. Furthermore, study introduces a hybrid framework that integrates feature engineering with a cuckoo-inspired optimisation model to improve prediction quality. This framework not only enhances predictive accuracy but also provides a more adaptable and generalizable approach for obesity risk prediction.

Despite extensive research on obesity prediction using machine learning, existing studies primarily focus on applying standard classifiers or isolated optimization techniques. Most works either emphasize algorithm comparison without systematic Hyperparameter optimization or apply feature selection independently of model tuning. As a result, many reported models suffer from limited generalizability, suboptimal performance, and insufficient interpretability. Furthermore, the combined impact of bio-inspired feature engineering and Hyperparameter optimization within a unified predictive framework remains underexplored in obesity risk assessment. To address these gaps, this study proposes a novel hybrid framework that integrates cuckoo-inspired feature optimization with systematic Hyperparameter tuning across multiple machine learning models. Unlike existing approaches, the proposed framework jointly optimizes feature relevance and model configuration, leading to enhanced accuracy, robustness, and interpretability. The study not only provides a comprehensive comparative evaluation of widely used classifiers but also demonstrates how nature-inspired optimization can significantly improve predictive reliability in preventive healthcare applications.

Main contributions of this work are as follows:

1. A comparative evaluation of multiple ML algorithms for obesity risk prediction.
2. Demonstration of critical role of Hyperparameter optimisation in improving accuracy, robustness, and generalisability of ML models.
3. Novel hybrid framework that combines feature engineering with a cuckoo-inspired optimisation algorithm to further improve predictive performance in healthcare applications.

By addressing dual challenges of feature refinement and model optimisation, this study adds to the growing body of research on ML in healthcare. Beyond its academic contributions findings have practical implications for developing preventive healthcare tools, supporting clinical decision-making and informing public health policies aimed at combating global obesity epidemic. Ultimately, study highlights transformative potential of machine learning in tackling one of today's most pressing health issues and emphasises need for continued advances in predictive modelling for preventive medicine.

Segmentation of paper is depicted as Section 2 Describes Literature Review, Section 3 Methodology in brief Result and Discussion in Section 4 and Finally Section 5 Concludes Conclusion and Future work in order to wrap research.

2. Literature review

Previous studies emphasize need for advanced analytical techniques beyond conventional statistical methods to predict obesity, given that it arises from complex interactions among environmental, behavioural and genetic factors [11]. Traditional approaches such as regression models and BMI often fail to capture these nonlinear relationships effectively. As a result use of machine learning (ML) in obesity research has expanded due to its ability to handle large, diverse datasets and uncover hidden patterns. Several widely used algorithms have been explored, including logistic regression, K-Nearest Neighbours, decision trees and support vector machines. Each offers unique strengths such as the interpretability of logistic regression, the strong classification capabilities of SVMs, and the simplicity of KNN yet they also present limitations, including overfitting, high computational cost and sensitivity to parameter. Literature consistently underscores the importance of Hyperparameter tuning in enhancing model accuracy and robustness. Common optimisation strategies include grid search, random search, Bayesian optimisation, and metaheuristic methods [12]. Moreover, feature engineering has proven to be a vital step in refining input data to improve predictive performance. More recent developments have introduced nature-inspired algorithms such as cuckoo search, which support both feature selection and parameter optimisation thereby producing more reliable models. While traditional ML algorithms remain effective for obesity risk prediction, their success largely depends on careful feature engineering and Hyperparameter optimisation. Hybrid approaches that integrate these techniques are particularly promising for creating more robust and accurate healthcare prediction systems. Based on the above analysis of existing studies, the proposed methodology is designed to overcome these limitations, as described in the following section. Recent studies have demonstrated the effectiveness of machine learning techniques for obesity risk prediction, with widely used classifiers such as Logistic Regression, Support Vector Machines, Decision Trees, and K-Nearest Neighbours showing improved performance over traditional statistical models. However, most existing works primarily focus on comparing algorithms under default or limited tuning conditions, which often leads to suboptimal generalization and reduced robustness. Some recent research has explored Hyperparameter optimization strategies and hybrid learning frameworks, indicating that model performance is highly sensitive to parameter configuration and feature relevance. Nevertheless, these approaches typically treat feature selection and Hyperparameter tuning as independent processes, and few studies integrate both within a unified predictive framework. Moreover, several high-performing models rely on complex ensemble or deep learning architectures that lack transparency, limiting their adoption in clinical and preventive healthcare settings. As a result, despite reported accuracy improvements, the novelty and practical advantages of many prior approaches remain unclear. Motivated by these limitations present

Table 1
Literature review of machine learning methods in obesity prediction.

Algorithms	Strengths	Limitations	Optimisation / Feature Engineering
Logistic Regression	High interpretability; simple implementation	Limited in capturing nonlinear relationships	Often improved with feature engineering
K-Nearest Neighbours (KNN)	Easy to implement; intuitive classification	Sensitive to choose of *k*; computationally expensive on large datasets	Performance improved via parameter tuning
Decision Trees	Handles nonlinear relationships; easy visualisation	Prone to overfitting; unstable with small data changes	Pruned or combined in ensembles
Support Vector Machines (SVM)	Strong classification performance; effective with high-dimensional data	High computational cost; sensitive to kernel and parameter selection	Grid search, random search used for tuning
Hyperparameter Optimisation	Grid search, random search, Bayesian optimisation, metaheuristic methods	Can be computationally demanding	Enhances robustness and accuracy
Feature Engineering	Improves predictive power; reduces noise in data	Requires domain knowledge; risk of overfitting	Feature scaling, selection, transformation
Nature-inspired Algorithms	Enhance feature selection and parameter tuning (e.g., cuckoo search)	Relatively new; computational complexity	Promising hybrid approaches

work proposes a novel hybrid framework that jointly integrates cuckoo-inspired feature optimization with systematic Hyperparameter tuning across multiple classical machine learning models. This unified optimization strategy enhances predictive accuracy, robustness, and interpretability simultaneously, while retaining computational efficiency and clinical transparency. By explicitly combining bio-inspired feature engineering, rigorous model optimization, and explainable analysis proposed approach addresses key gaps in existing literature and demonstrates clear advantages over prior obesity risk prediction studies.

Table 1 shows Literature Review.

3. Methodology

Fig. 1 presents the overall workflow of the proposed machine learning–based obesity risk prediction framework. The process begins with data collection from a publicly available dataset, followed by data Preprocessing steps including missing value handling, normalization, and categorical encoding. The dataset is then divided into training and testing subsets. Feature engineering is performed using a cuckoo-

Table 2
Attribute discerptions and range.

Attribute Name	Description	Range / Possible Values
Gender	Biological sex of the individual	Male / Female
Age	Age of the person in years	14 – 61
Height	Height of the person in meters	1.45 – 1.98
Weight	Weight of the person in kilograms	39 – 173
Family History of Overweight	Whether any family member is overweight	Yes / No
FAVC	Frequent consumption of high-calorie food	Yes / No
FCVC	Frequency of vegetable consumption	1 – 3 (1: Never, 2: Sometimes, 3: Always)
NCP	Number of main meals per day	1 – 4
CAEC	Consumption of food between meals	No / Sometimes / Frequently / Always
SMOKE	Smoking habit	Yes / No
CH2O	Daily water intake (in litres)	1 – 3 (1: Less than 1 L, 2: 1–2 L, 3: More than 2 L)
SCC	Monitors calorie consumption	Yes / No
FAF	Frequency of physical activity	0 – 3 (0: None, 3: High activity)
TUE	Time spent using technology devices (hours/day)	0 – 2 (0: <2 hrs, 1: 3–5 hrs, 2: >5 hrs)
CALC	Frequency of alcohol consumption	No / Sometimes / Frequently / Always
MTRANS	Mode of transportation	Automobile / Motorbike / Bike / Public Transport / Walking
Obesity (Target)	Obesity level classification	Underweight,Normal / Overweight / Obesity I / Obesity II / Obesity III

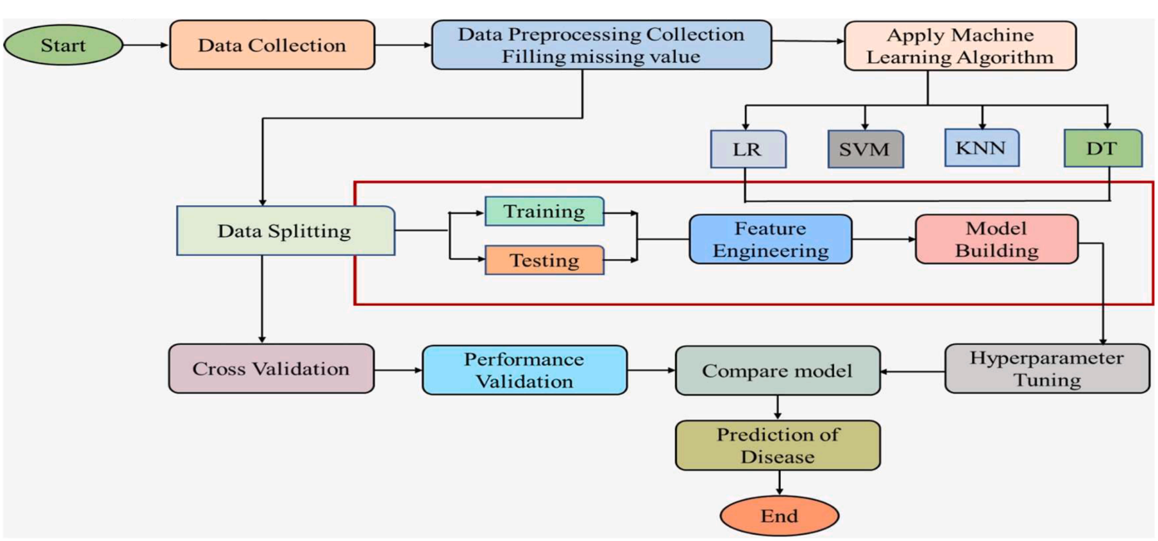

Fig. 1. Proposed frame work.

Table 3
Data pre-processing techniques applied.

Attribute	Type	Pre-processing Technique Applied	Remarks
Gender	Categorical	Label Encoding	Converted Male/ Female to binary values
Age	Numerical	Normalization (Min–Max Scaling)	Scaled to 0–1 range
Height	Numerical	Normalization (Min–Max Scaling)	Standardized to uniform scale
Weight	Numerical	Normalization (Min–Max Scaling)	Reduced skewness
Family History of Overweight	Categorical	Label Encoding	Encoded Yes/No to binary values
FAVC	Categorical	Mode Imputation, Label Encoding	Replaced missing values with mode
FCVC	Numerical	Normalization	Values scaled between 0–1
NCP	Numerical	Normalization	Continuous variable standardized
CAEC	Categorical	One-Hot Encoding	Transformed categorical frequency levels
SMOKE	Categorical	Mode Imputation, Label Encoding	Binary encoding applied
CH2O	Numerical	Normalization	Scaled to uniform range
SCC	Categorical	Label Encoding	Encoded Yes/No as 0/1
FAF	Numerical	Normalization	Scaled 0–1 for uniformity
TUE	Numerical	Normalization	Device usage normalized
CALC	Categorical	One-Hot Encoding	Frequency-based categorical variable
MTRANS	Categorical	One-Hot Encoding	Multi-class variable encoded
Obesity (Target)	Categorical	Label Encoding	Converted class labels into numerical values

inspired optimization algorithm to identify the most relevant attributes. Subsequently, multiple machine learning models Logistic Regression, Support Vector Machine, K-Nearest Neighbour, and Decision Tree are trained using the optimized feature set [13]. Hyperparameter tuning is applied to improve generalization and predictive performance. Finally, the models are evaluated using standard performance metrics, and the best-performing model is selected for obesity risk prediction.

3.1. Data collection

dataset used in this study was obtained from UCI Machine Learning Repository and comprises 17 attributes related to obesity. These attributes encompass demographic, behavioural and physical factors such as age, gender, height, weight, BMI, dietary habits, level of physical activity and other lifestyle variables. Containing 2110 records with data values ranging from 0 to 2110 dataset represents a diverse set of individual profiles. It provides a comprehensive overview of factors influencing obesity, making it suitable for training and evaluating various ML models [14]. By applying feature engineering and Hyperparameter tuning this dataset enables effective comparison and optimization of multiple algorithms, serving as the foundation for developing and validating proposed obesity prediction framework as shown in Table 2.

3.2. Data pre-processing and missing values

Data Pre-processing is one of most essential steps to ensure that dataset is clean, consistent and suitable for effective model training. Obesity dataset used in this study obtained from UCI Machine Learning Repository, underwent several Pre-processing stages to enhance its quality and analytical reliability. Process began with checking dataset for missing or inconsistent values. Since dataset contained both numerical and categorical variables, mean or median imputation was applied to continuous attributes such as age, height, and weight, while mode imputation was used for categorical variables including gender, FAVC, and SMOKE. Duplicate entries and irrelevant attributes were removed to maintain data integrity and ensure dataset's accuracy. Subsequently, data normalization was performed on numerical features to guarantee that each variable contributed equally during model training. Using Min–Max scaling technique data values were rescaled within a standard range of 0 to 1, thereby minimizing dominance of features with larger magnitudes. To make categorical variables compatible with MLM label encoding and one-hot encoding were applied as appropriate. Pre-processed dataset was then divided into training (80 %) and testing (20 %) subsets to evaluate model performance objectively [15]. Through these pre-processing procedures, dataset was refined to enhance precision, reduce noise and provide reliable input for subsequent feature engineering and model training stages. A summary of pre-processing techniques applied to each variable is presented in Table 3.

3.3. Data splitting

pre-processed obesity dataset was subsequently divided into training and testing subsets to evaluate performance and generalization ability of MLM. Data splitting is a crucial step that ensures models are trained on one portion of data and tested on unseen samples, thereby preventing overfitting and improving predictive reliability [16]. In this study 80:20 ratio was employed with 80 % of dataset allocated for training and 20 % reserved for testing. Training set was used for model development and optimization through feature engineering and Hyperparameter tuning while testing set was utilized to assess models predictive accuracy on new data. This division enabled an objective evaluation of each algorithm's accuracy, precision, recall, and F1-score in predicting obesity levels. To ensure consistency and reproducibility of results across multiple experimental runs a fixed random state parameter was applied.

3.3.1. Cross validation

To ensure robustness and reliability of developed MLM cross-validation was employed during model evaluation phase. This technique helps prevent overfitting and provides a more accurate estimate of a model's predictive performance by testing its ability to generalize to unseen data. In this study, k-fold cross-validation with $k = 10$ was applied, dividing dataset into ten approximately equal folds. During each iteration, nine folds were used for training while the remaining fold was used for validation, and this process was repeated until every fold had served as the validation set once. Results from all iterations were then averaged to compute the final performance metrics, including accuracy, precision, recall, and F1-score. Models such as Logistic Regression (LR), Support Vector Machine (SVM), K-Nearest Neighbour (KNN), and Decision Tree (DT) were systematically evaluated to ensure that their performance was not biased toward any particular subset of data. By applying cross-validation, the study achieved consistent and dependable results, establishing a strong foundation for comparing the predictive capabilities of the different algorithms [17,18].

3.4. Machine learning algorithms

ML helps predict and categorize obesity risk by analysing health factors DT model makes predictions through a hierarchy of decision rules KNN) classifies individuals based on similar profiles, Support Vector Machine (SVM) separates obese and non-obese groups using optimal decision boundaries, and Logistic Regression (LR) estimates the probability of obesity. Together, these algorithms enable early obesity detection and provide personalized health recommendations.

LR: it is data analysis technique that uses mathematical methods to identify relationships between variables. This machine learning approach is popular because it is fast and delivers high accuracy.

Algorithm 1
Cuckoo search–based feature engineering for obesity prediction.

Input: Dataset D, ML model M, population size N, discovery rate Pα max iterations MaxIter

1. **Initialize** N nests with random feature subsets.
2. **Evaluate fitness** of each nest using M (e.g., accuracy).
3. For each iteration:
   a. Generate new feature subset using Lévy flight.
   b. Evaluate its fitness.
   c. Replace a random nest if new fitness is better.
   d. Abandon fraction Pα of worst nests and create new ones.
   e. Retain the best nest.
4. **Select** best feature subset Fopt from highest-fitness nest.
5. **Retrain** ML model M using Fopt and record performance.

Output: Optimal features Foptand final model accuracy.

Although it resembles a linear regression model, LR is specifically designed for categorical dependent variables estimating probability that a given input belongs to a particular class. It can also be extended to handle multiple classes through methods such as One-vs-All (OvA) and Softmax regression. LR is especially useful when interpretable results are needed and the relationship between input features and target variable is approximately linear [19].

SVM: SVM model distinguishes between obese and non-obese individuals by finding optimal boundary (hyperplane) in feature space that separates the two groups based on factors like BMI, diet, and activity level. Kernel functions enhance its ability to handle both linear and nonlinear data effectively [20,21].

$$F(x)=W^{T}_{X+B} \tag{1}$$

w = weight vector (defines the hyperplane direction) x = input features (e.g., age, BMI, calories, exercise) b = bias term

KNN: algorithm is a data classification technique that predicts the likelihood of a data point belonging to a particular group based on its proximity to other points. KNN is a supervised machine learning method that can be applied to both regression and classification tasks, though it is mainly used for classification. Unlike other algorithms, KNN does not perform explicit computation or training on the dataset. Instead, it determines the class of a new data point by analysing which group its nearest neighbours belong to such as deciding whether it falls into group A or group B [22].

DT: Decision Tree classifiers belong to the family of supervised learning algorithms used for both classification and regression tasks. They use a tree-like structure where each internal node represents a decision based on a feature, and each leaf node corresponds to a class label or outcome. Entire training dataset starts at the root node, and statistical methods are applied to determine the feature that best splits the data at each internal node for effective classification [23].

3.5. Feature engineering

To enhance predictive performance of machine learning (ML) models in obesity research feature engineering plays a vital role. Because obesity arises from complex interactions among genetic, behavioural, environmental and physiological factors, raw datasets often include redundant or irrelevant variables that can degrade model accuracy. Effective feature engineering transforms these raw inputs into meaningful, high-quality features that capture underlying dynamics of health. Cuckoo Search (CS) algorithm a metaheuristic optimization technique inspired by brood parasitic behaviour of certain cuckoo species is widely used for feature selection and dimensionality reduction. In context of obesity prediction Cuckoo model efficiently identifies optimal subsets of features that contribute most to accurate classification or regression outcomes while filtering out noisy or redundant data. Each nest in algorithm represents a potential feature subset, and new candidate solutions are generated through Lévy flights to explore global feature space [24]. Underperforming nests are replaced ensuring

Table 4
Grid and random search hyperparameter parameters.

Model	Hyperparameter	Values Considered
Logistic Regression (LR)	Regularization Strength (C) Solver Penalty	[0.01, 0.1, 1, 10, 100] ['liblinear', 'lbfgs', 'saga'] ['l1', 'l2']
Support Vector Machine (SVM)	Kernel Regularization (C) Gamma	['linear', 'poly', 'rbf', 'sigmoid'] [0.1, 1, 10, 100] ['scale', 'auto']
K-Nearest Neighbour (KNN)	Number of Neighbours (k) Distance Metric Weight Function	[3, 5, 7, 9, 11] ['euclidean', 'manhattan', 'minkowski'] ['uniform', 'distance']
Decision Tree (DT)	Criterion Max Depth Min Samples Split Min Samples Leaf	['gini', 'entropy'] [None, 5, 10, 20] [2, 5, 10] [1, 2, 4]

convergence toward feature combinations that balance exploration and exploitation.

When integrated with machine learning algorithms such as Support Vector Machines (SVM), Logistic Regression (LR), Decision Trees (DT) or K-Nearest Neighbors (KNN), Cuckoo-based feature engineering substantially enhances model interpretability, reduces computational cost, and minimizes overfitting by focusing on most informative feature such as dietary habits, physical activity levels, metabolic indicators and demographic attributes. Empirical studies show that Cuckoo Search based feature selection outperforms traditional methods like Principal Component Analysis (PCA) and Recursive Feature Elimination (RFE particularly for nonlinear and high-dimensional obesity datasets. This hybrid approach, combining bio-inspired optimization with machine learning, provides a robust framework for accurate obesity prediction, early risk detection and personalized intervention strategies [25]. Overall, Cuckoo model driven feature engineering offers an intelligent and powerful method for managing complex health data and advancing precision medicine in obesity research and Algorithm is given below; (Algorithm 1)

3.6. Hyperparameter tuning

Hyperparameter tuning was a key aspect of this study, aimed at maximizing predictive performance across all the machine learning

Table 5
Best parameters for different algorithms.

Model	Best Parameters Identified
Logistic Regression (LR)	C = 1.0; Solver = 'lbfgs'; Penalty = 'l2'
Support Vector Machine (SVM)	Kernel = 'rbf'; C = 10; Gamma = 'scale'
K-Nearest Neighbour (KNN)	n_neighbors = 5; Metric = 'minkowski'; Weights = 'distance'
Decision Tree (DT)	Criterion = 'gini'; Max Depth = 10; Min Samples Split = 2; Min Samples Leaf = 1

Table 6
Performance comparison of machine learning models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Logistic Regression (LR)	99.63	99.40	99.50	99.45
K-Nearest Neighbour (KNN)	94.33	94.10	94.25	94.18
Support Vector Machine (SVM)	89.60	89.20	89.35	89.27
Decision Tree (DT)	79.68	79.00	79.30	79.15

algorithms explored. It is a crucial step in optimizing model configurations to achieve the best possible results. For each algorithm, the optimal parameter combinations were identified by systematically adjusting factors such as learning rate, regularization strength, and tree depth [26]. process involved an extensive search of Hyperparameter space using techniques like grid search and randomized search to enhance model generalization and prediction accuracy across multiple datasets. Table 4 presents specific Hyperparameter values considered for each model during these optimization procedures. This table provides an overview of range of Hyperparameter that were evaluated and highlights the tuning process that led to identification of optimal parameters presented in Table 5.

3.7. Performance validation

performance validation process employed different evaluation metrics accuracy, precision, recall and F1-score to assess effectiveness and predictive capability of developed MLM. These metrics provide a comprehensive evaluation of each algorithms ability to classify obesity levels accurately and generalize to unseen data. Accuracy represents overall proportion of correctly predicted instances while precision measures percentage of correctly identified positive cases among all predicted positives, reflecting model's reliability in detecting true obesity cases. Recall also known as sensitivity, evaluates model's ability to identify all actual positive cases, indicating how effectively it captures individuals with obesity. F1-score which is harmonic mean of precision and recall provides a balanced measure of both metrics, particularly useful when working with imbalanced datasets.

By comparing these performance indicators across models such as DT, KNN, SVM and LR study ensured a fair and comprehensive evaluation of predictive performance. This validation process helped identify the most accurate and reliable model for obesity prediction, confirming that selected model not only performed well on training data but also demonstrated strong generalization on testing dataset. Table 6 summarizing performance validation.

4. Result and discussion

In this section different machine learning algorithms DT, SVM, KNN and LR were evaluated for obesity risk prediction using a dataset obtained from the UCI repository. The models were assessed based on key performance metrics: accuracy, precision, recall, and F1-score. Among algorithms tested, Logistic Regression achieved the highest accuracy of 99.63 %, demonstrating strong generalization and reliability in identifying obesity levels. KNN followed with an accuracy of 94.33 %, effectively recognizing patterns through similarity-based classification. SVM achieved 89.6 % accuracy, showing reasonable performance though slightly reduced sensitivity due to kernel parameter dependence. Decision Tree, while easy to interpret, recorded the lowest accuracy of 79.68 %, likely caused by overfitting and sensitivity to minor data variations. Results clearly indicate that Hyperparameter tuning significantly enhanced model performance, improving both accuracy and robustness across all algorithms. Regularization adjustments most benefited LR, while optimal kernel and gamma configurations enhanced SVM outcomes. KNN performance improved by fine-tuning number of neighbours and distance measures, and DT stability increased after pruning. Overall, findings highlight the crucial role of feature engineering and Hyperparameter optimization in developing reliable obesity prediction systems. Logistic Regression proved to be most effective model, balancing interpretability with predictive strength. Additionally, incorporating advanced feature selection techniques such as cuckoo-inspired optimization model further improved performance and minimized overfitting, supporting creation of accurate, data-driven tools for early obesity risk detection and preventive healthcare applications. In Fig. 2 All attributes are grouped within their respective value ranges and displayed as normally distributed histograms. X-axis represents type or nature of each attribute while y-axis shows corresponding value.

In Fig. 3 Heat map presents pairwise correlation coefficients among variables in obesity dataset. Color gradient depicts direction and strength of linear relationships with darker shades representing negative correlations and lighter shades indicating stronger positive correlations.

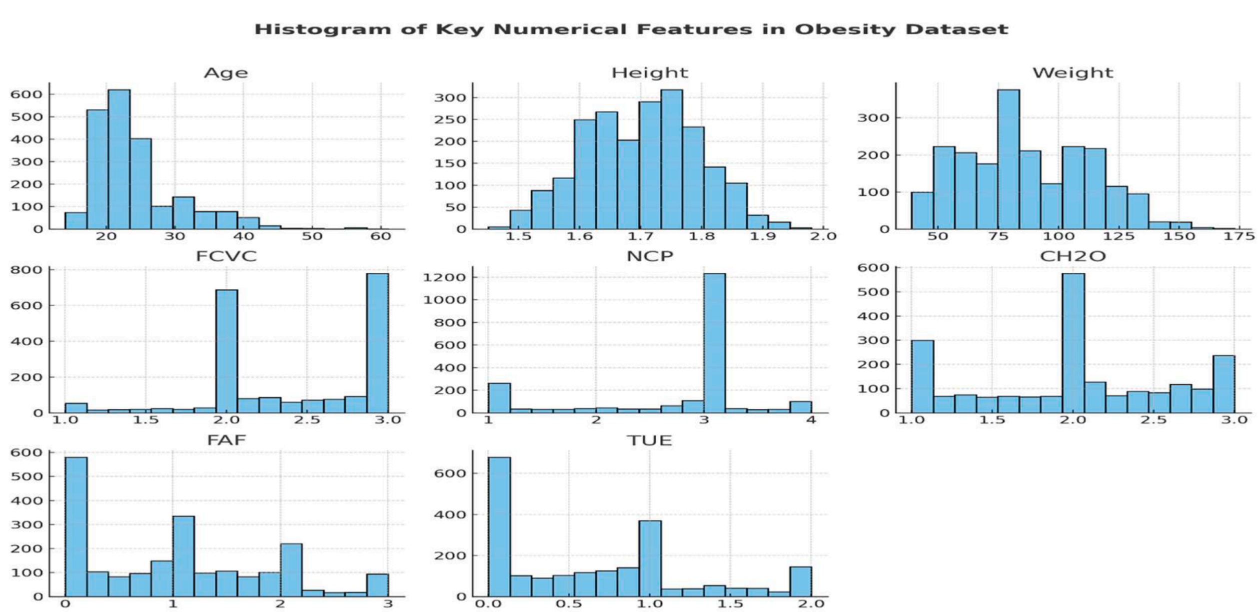

Fig. 2. Histogram of Obesity data set.

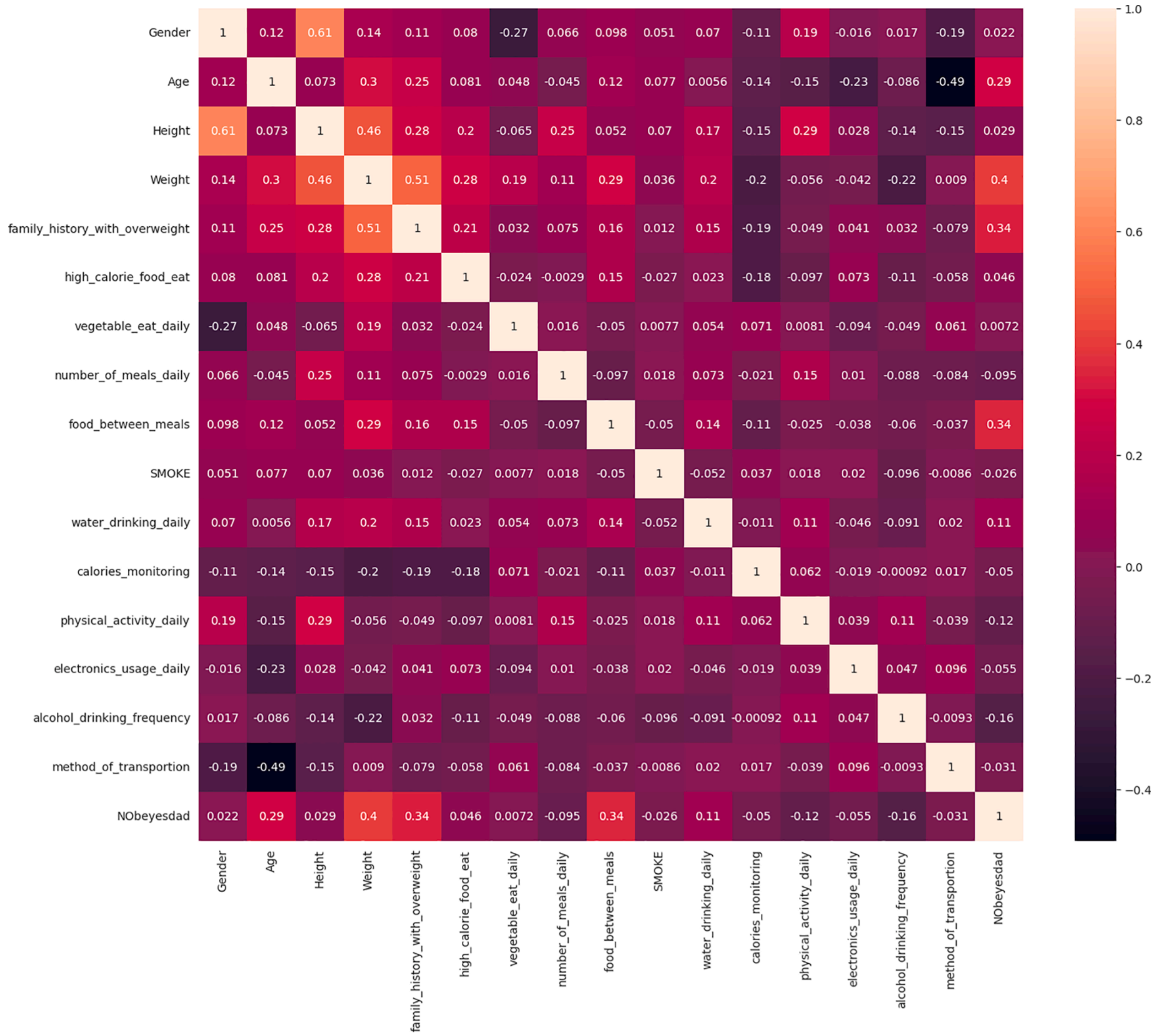

Fig. 3. Inter-Attribute Correlation Analysis of Obesity Dataset.

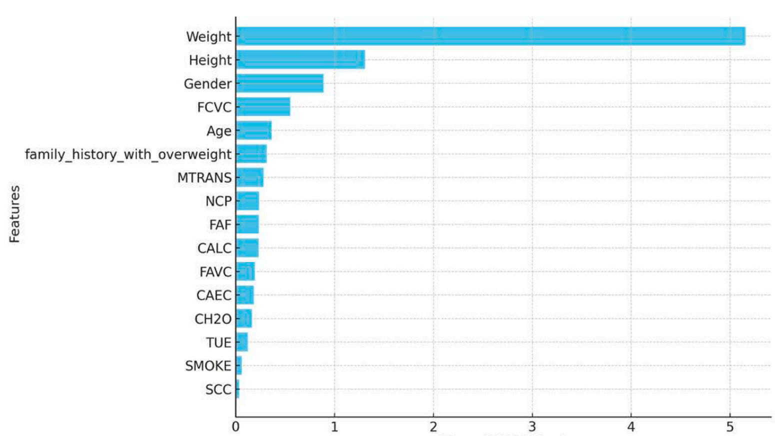

Fig. 4. Feature importance of SHAP Plot.

Notably, Age shows weak correlations with most attributes, whereas Height and Weight display a moderately positive relationship. Obesity level is moderately correlated with both Weight and family_history_with_overweight, suggesting that these factors may influence obesity outcomes. Overall, matrix provides valuable insight into physiological, behavioural and lifestyle factors contributing to obesity.

Fig. 4. SHAP feature importance derived from the Logistic Regression model across all obesity classifications. Results indicate that weight, height, and gender are most influential factors in determining obesity level followed by age and FCVC (food consumption). Other attributes such as family history of obesity and mode of transportation, exhibit a moderate to minor impact.

4.1. Assumptions and limitations

This study assumes that the UCI obesity dataset is representative of the target population and that the selected demographic and lifestyle features are sufficient for accurate obesity risk prediction. It also assumes data consistency between training and testing phases and effective convergence of the optimization techniques. However, study is limited by reliance on a single cross-sectional dataset and the absence of genetic, biochemical, and longitudinal health data. The computational cost of bio-inspired optimization and restriction to classical machine learning models may further limit scalability and generalizability to real-world clinical.

5. Conclusion and future work

This study presented a comprehensive machine learning–based framework for obesity risk prediction by jointly integrating bio-inspired feature engineering and systematic Hyperparameter optimization. A key strength of the proposed research lies in its unified approach, where feature relevance and model configuration are optimized simultaneously rather than independently. This strategy significantly enhanced predictive accuracy, robustness, and interpretability across multiple classifiers. The extensive comparative evaluation demonstrated that Logistic Regression, when combined with optimized features and tuned parameters, achieved superior performance (99.63 % accuracy), offering an effective balance between high predictive power and clinical interpretability. The inclusion of SHAP-based explain ability further strengthens the study by providing transparent insights into influential risk factors such as weight, height, gender, age, and dietary behavior, which is critical for healthcare adoption.Despite these strengths, the study has certain limitations. The experiments were conducted using a single publicly available, cross-sectional dataset, which may limit generalizability across diverse populations. Additionally, the framework focuses on classical machine learning models, and the computational overhead introduced by bio-inspired optimization techniques may pose scalability challenges for very large datasets or real-time systems. Assessment of results indicates that Hyperparameter tuning and optimized feature selection play a decisive role in improving obesity risk prediction accuracy and stability. The findings reinforce the importance of systematic optimization in healthcare machine learning applications and demonstrate that interpretable models, when properly optimized, can outperform more complex alternatives. From a broader perspective, the results have meaningful implications for preventive healthcare, as they support early identification of high-risk individuals and facilitate evidence-based decision-making for clinicians and policymakers. In terms of applications and recommendations, the proposed framework can be effectively integrated into clinical decision support systems, digital health platforms, and community-level screening tools for early obesity risk assessment. It may also assist public health authorities in designing targeted intervention strategies. Future work should focus on incorporating longitudinal, genetic, and biochemical data to further enhance predictive reliability. Additionally, extending the framework to ensemble and deep learning models, as well as deploying it in real-time mobile or web-based health monitoring systems, represents a promising direction for advancing personalized and preventive obesity management.

CRediT authorship contribution statement

Salliah Shafi: Writing – original draft, Conceptualization. **Gufran Ahmad Ansari:** Writing – review & editing, Data curation, Conceptualization. **Lamees Alhazzaa:** Validation, Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this research paper.

Data availability

The data that has been used is confidential.

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